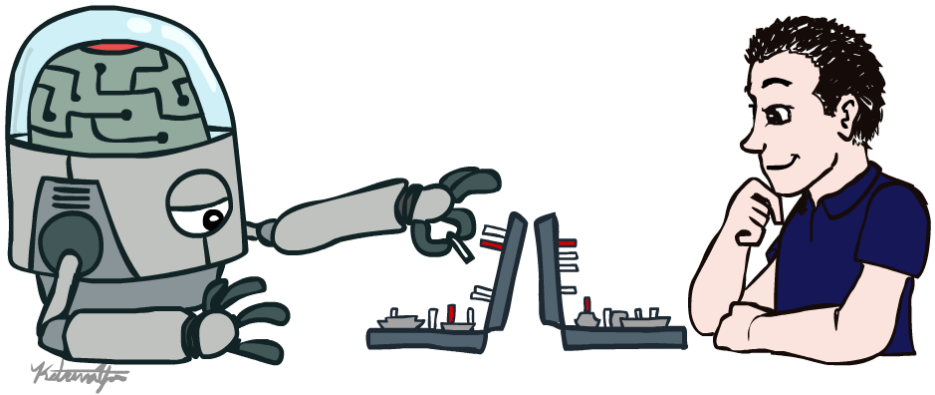
Osnovi Računarske Inteligencije

Uvod



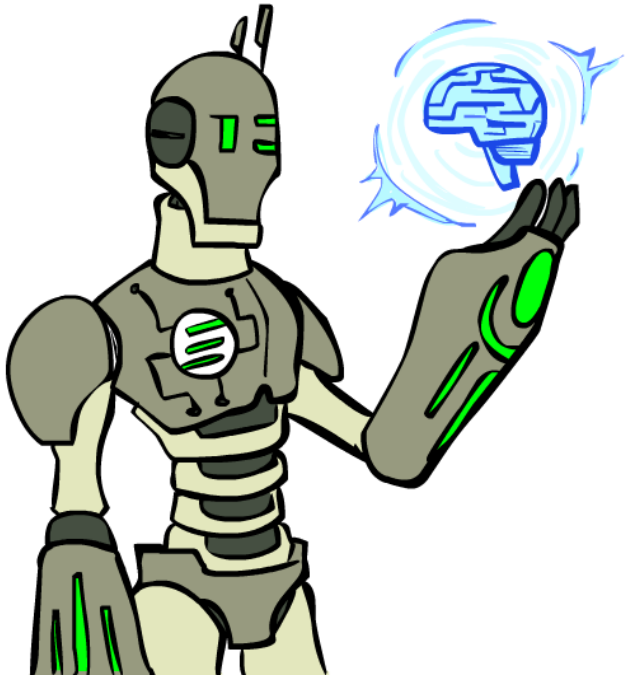
Predavač: Aleksandar Kovačević

Slajdovi preuzeti sa kursa CS188, University of California, Berkeley

http://ai.berkeley.edu/

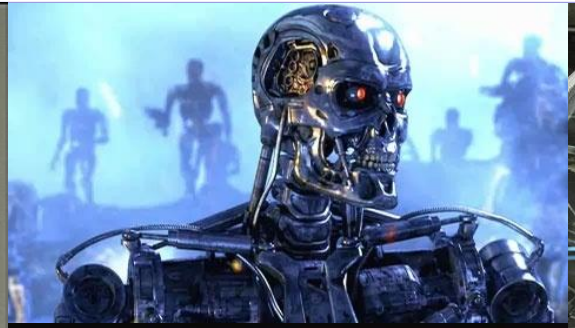
Danas

- Šta je veštačka inteligencija?
- Šta AI može da uradi?
- Šta učimo na ovom kursu?

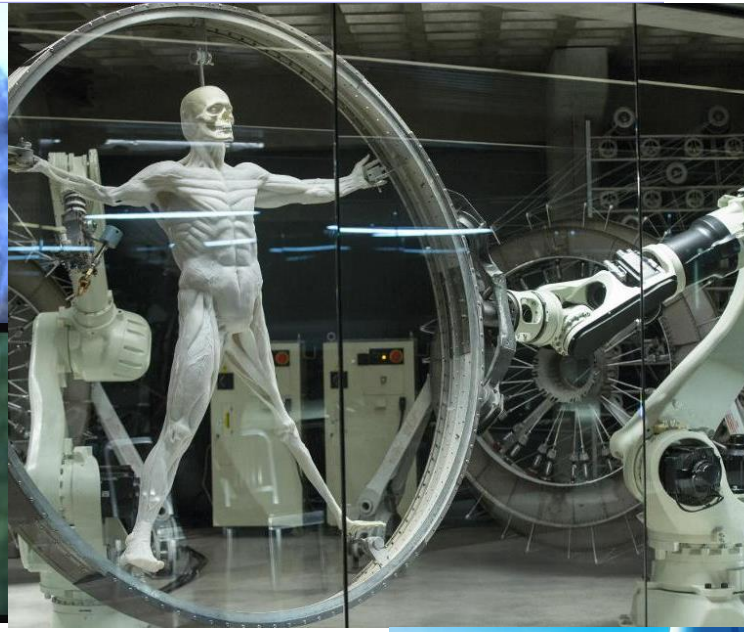


AI u filmovima











AI u vestima

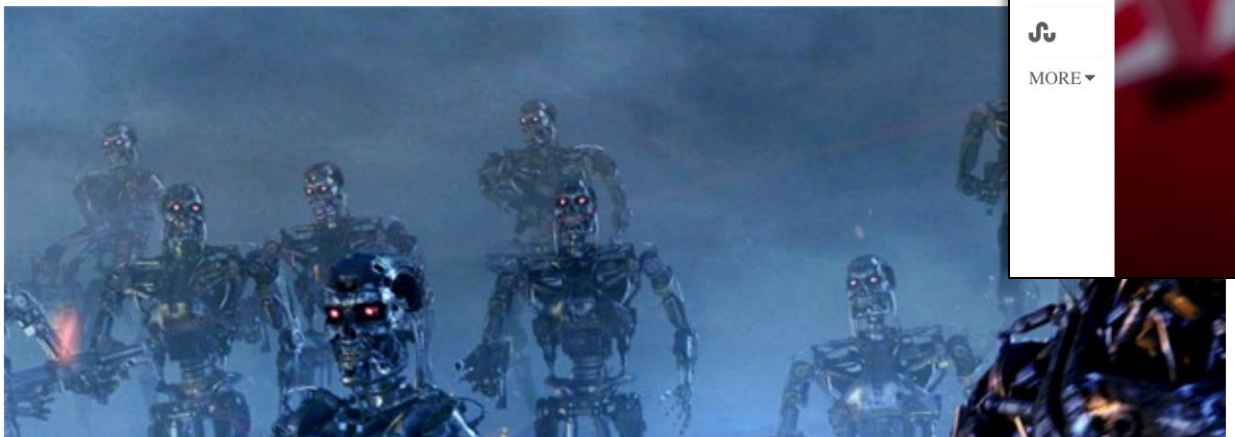
AI is the biggest risk we face as a civilisation, Elon Musk says

Billionaire burn: Musk says Zuckerberg's understanding of AI threat 'is limited'

HOME » FINANCE » FINANCE TOPICS » DAVOS

'Sociopathic' robots could overrun the human ra a generation

Computers should be trained to serve humans to reduce their th human race, says a leading expert on artificial intelligence



LIVESCIENCE NEWS TECH HEALTH PLANET EAR

Live Science › Tech

Lifelike 'Sophia' Robot Granted Citizenship to Saudi Arabia

By Mindy Weisberger, Senior Writer | October 30, 2017 03:39pm ET

Be the most interesting person you know, subscribe to LiveScience. Subscribe ›



AI u vestima

TECH • ARTIFICIAL INTELLIGENCE

United Kingdom Plans $1.3 Billion Intelligence Push

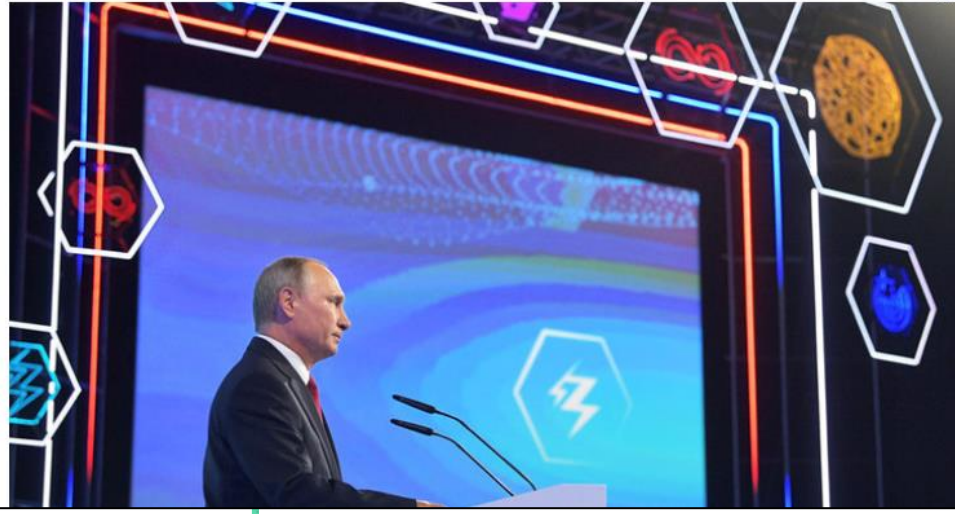
France to spend $1.8 billion on compete with U.S., China

EU wants to invest £18b development

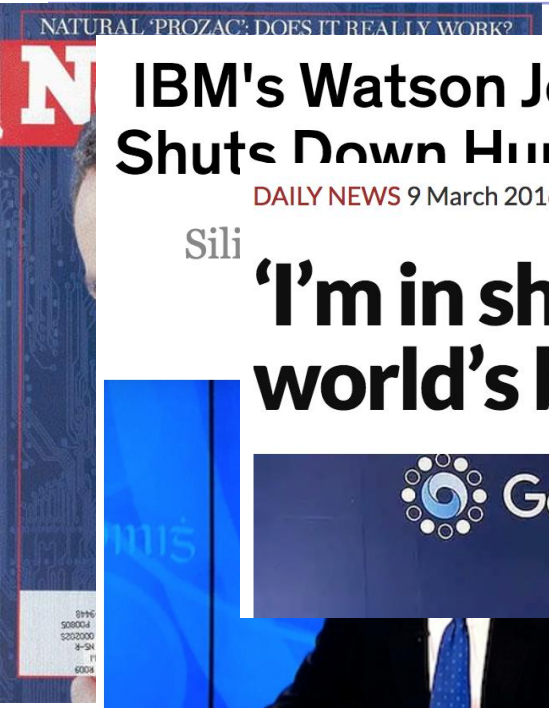
China's Got a Huge Art Intelligence Plan

'Whoever leads in AI will rule the world': Putin to Russian children on Knowledge Day

Published time: 1 Sep, 2017 14:08
Edited time: 1 Sep, 2017 14:40



AI u vestima

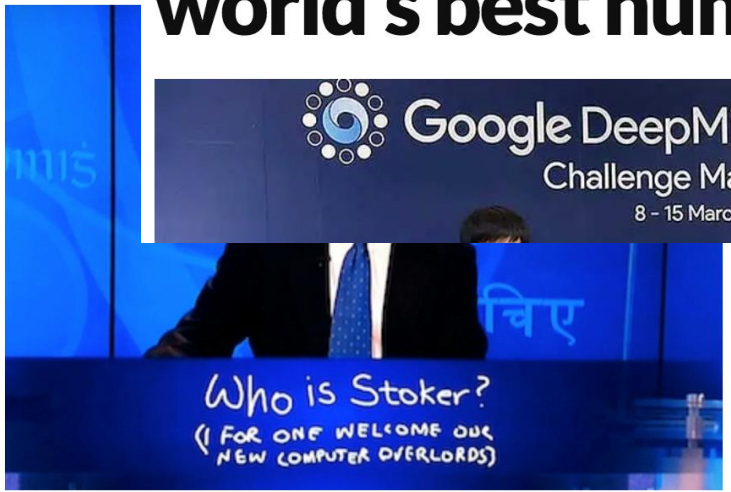


IBM's Watson Jeopardy Computer Shuts Down Humans in Final Game

DAILY NEWS 9 March 2016

Sili

‘I’m in shock!’ Ho world’s best hum



Blizzard will show off Google's Deepmind AI in StarCraft 2 later this week

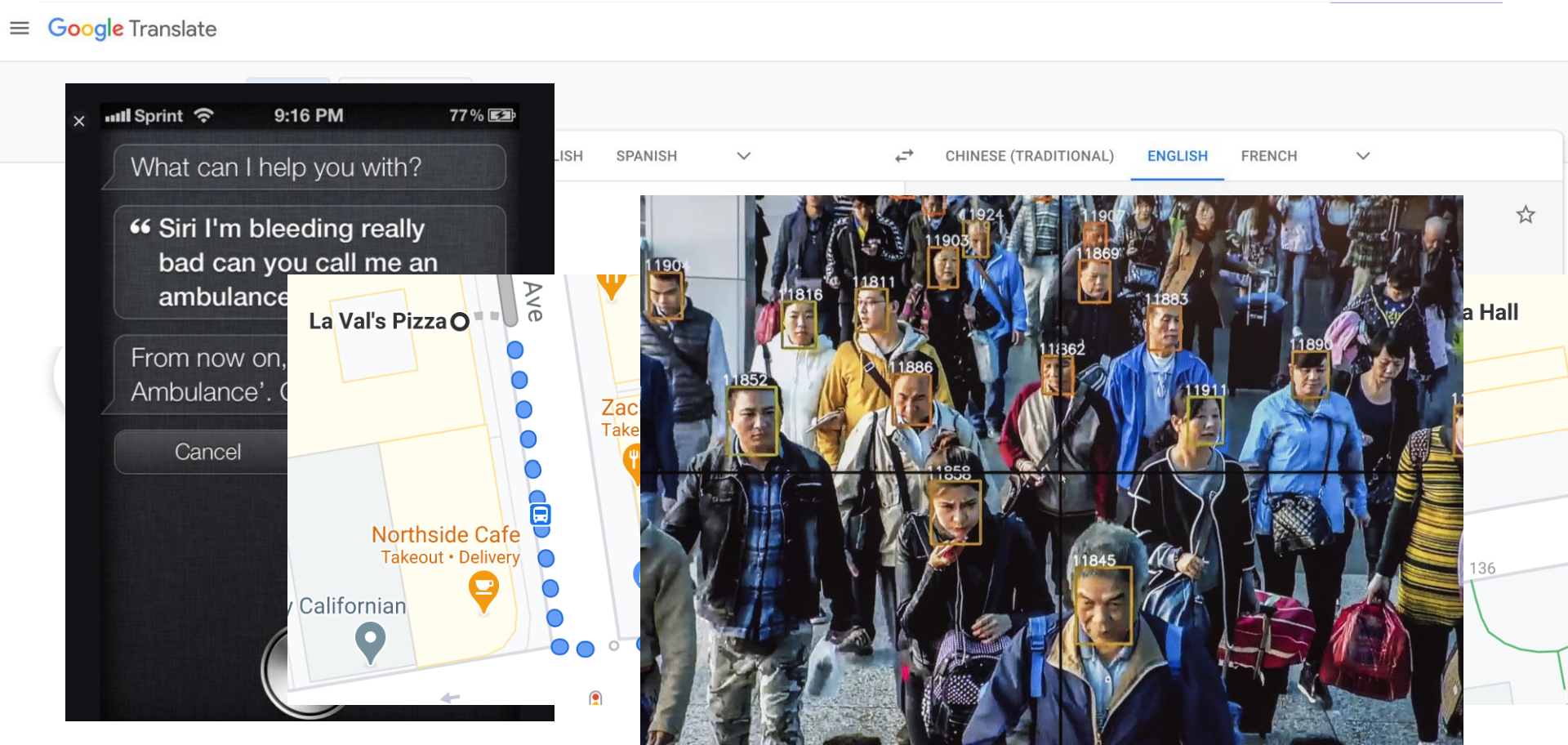
By Andy Chalk 4 hours ago

Google and Blizzard launched the artificial intelligence project in 2016.

COMMENTS



AI u realnosti




TUG
CAUTION
MAY CONTAIN
CHEMOTHERAPY DRUG
CAUTION
MAY CONTAIN
CHEMOTHERAPY DRUG

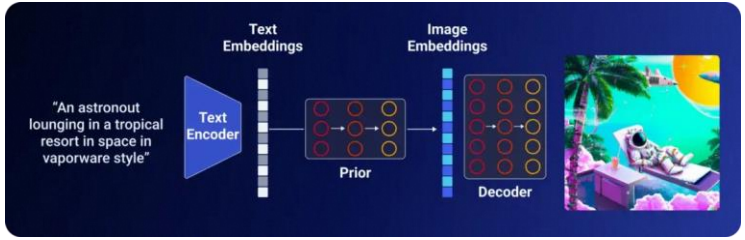



Boston Dynamics



AI u 2022. godini – DALL·E 2









AI u 2022. godini - ChatGPT

OpenAI

CHATGPT

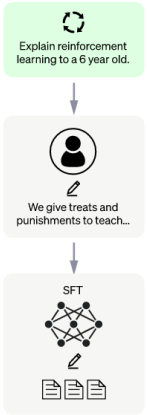
Step 1

Collect demonstration data and train a supervised policy.

A prompt is sampled from our prompt dataset.

A labeler demonstrates the desired output behavior.

This data is used to fine-tune GPT-3.5 with supervised learning.



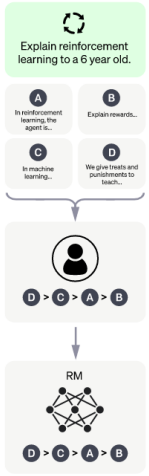
Step 2

Collect comparison data and train a reward model.

A prompt and several model outputs are sampled.

A labeler ranks the outputs from best to worst.

This data is used to train our reward model.



Step 3

Optimize a policy against the reward model using the PPO reinforcement learning algorithm.

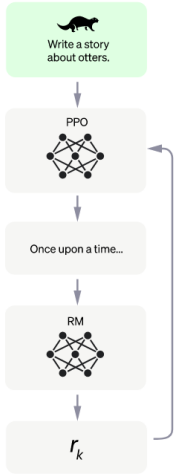
A new prompt is sampled from the dataset.

The PPO model is initialized from the supervised policy.

The policy generates an output.

The reward model calculates a reward for the output.

The reward is used to update the policy using PPO.



Give a step-by-step analysis of the worst-case time complexity of the bubble sort algorithm, with Python code examples, but write every sentence in the speaking style of a fast-talkin' wise guy from a 1940's gangster movie.

Alright, pal, listen up. The bubble sort algorithm, she's a real simple one. But the worst-case time complexity, that's a whole different story. You see, in the worst case, every element in the list gotta be swapped, so the time complexity is gonna be proportional to the square of the length of the list. Get it?

Here's some Python code to help you out, see if you can follow along.

Copy code

```
# Start with a list of numbers that ain't sorted
numbers = [5, 1, 4, 2, 8]

# Keep track of whether any swaps were made on the previous iteration
# If no swaps were made, the list is already sorted and we can stop
swapped = True

while swapped:
    # Set swapped to False so we can check if any swaps are made later
    swapped = False

    # For each pair of adjacent elements,
    for i in range(len(numbers) - 1):
        # If the first element is greater than the second,
        if numbers[i] > numbers[i + 1]:
            # Swap the elements
            numbers[i], numbers[i + 1] = numbers[i + 1], numbers[i]
            # Set swapped to True so we know to keep looping
            swapped = True

# At this point, the list is sorted
print(numbers)
```

See, in the worst case, the while loop is gonna keep loopin' until there ain't no more swaps to be made. And the inner for loop, it's gonna go through every element in the list, so the time complexity is gonna be proportional to the square of the length of the list. Get it?

Šta je AI?

Oblast koja se bavi stvaranjem mašina koje:

Racionalne Odluke

Termin **racionalnost** za nas ima vrlo specifičnu tehničku definiciju:

- Racionalnost: obavljanje unapred definisanih ciljeva na najbolji mogući način.
- Zanima nas samo koje će odluke AI doneti (ne i način na koji ih je doneo).
- Ciljeve definišemo pomoću **korisnosti** (***utility***) tj. kvaliteta ishoda koji je rezultat niza odluka AI.
- Biti racionalan za nas znači **maksimizovati svoju očekivanu korisnost** (***maximizing your expected utility***).

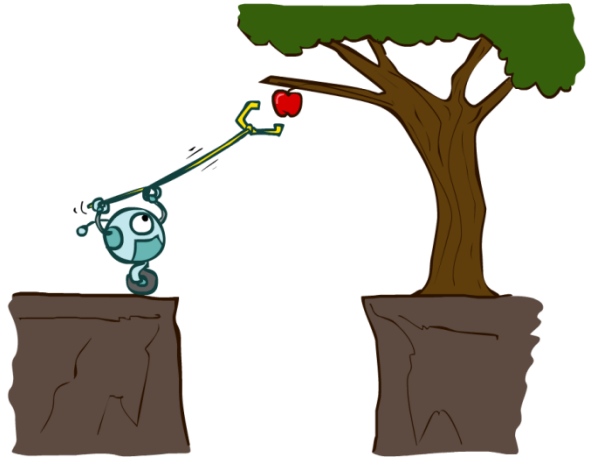
Šta možemo da naučimo proučavanjem mozga?

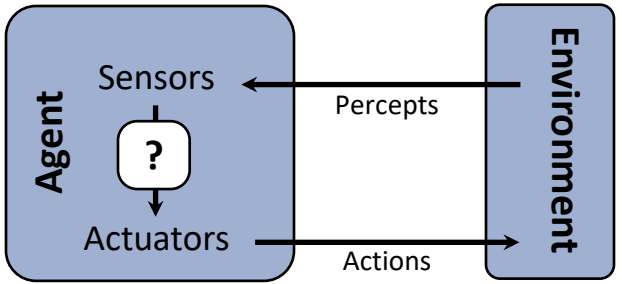
- Ljudski mozgovi su jako dobri u donošenju racionalnih odluka, ali nisu savršeni.
- Mozak nije baš modularan kao sofver, reverse engineering mozga je jako težak!
- Za AI oblast: “Veza mozga i inteligencije je kao veza krila i letenja”
- Šta je do sada naučeno proučavanjem mozga: pamćenje i simulacija su ključni za donošenje odluka.



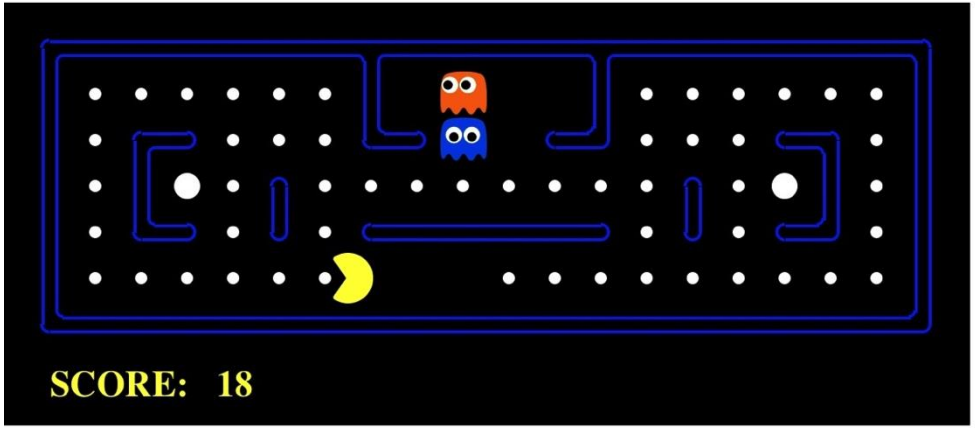
Kreiranje Racionalnih Agenata

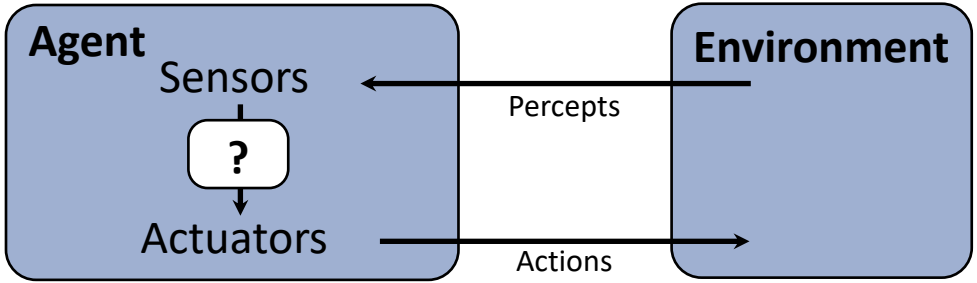
- **Agent** je entitet koji *sagledava (percepira) okruženje* i *deluje*.
- **Racionalan agent** bira svoje akcije tako da maksimizuje svoju (očekivanu) **korisnost**.
- Karakteristike **okoline**, načina **percepcije** i **mogućih akcija** određuju koja će tehnika biti upotrebljena za donošenje racionalnih odluka.
- **Ovaj kurs je o**:
  - Učenju tehnika veštačke inteligencije vezanih za rešavanje puno različitih vrsta problema
  - Učenju o tome koje tehnike treba odabrati za rešavanje novog problema koji se pojavio.



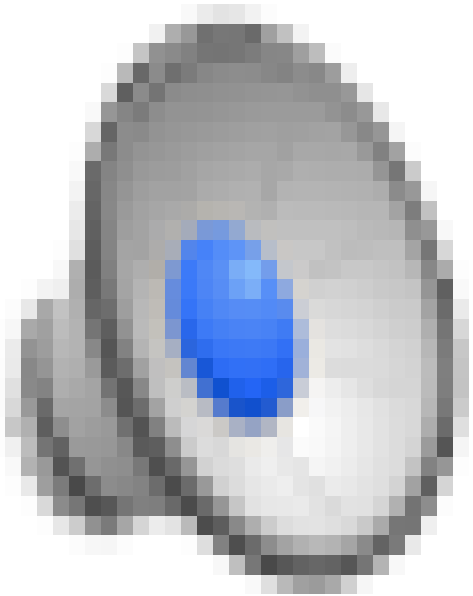


Pac-Man kao Agent





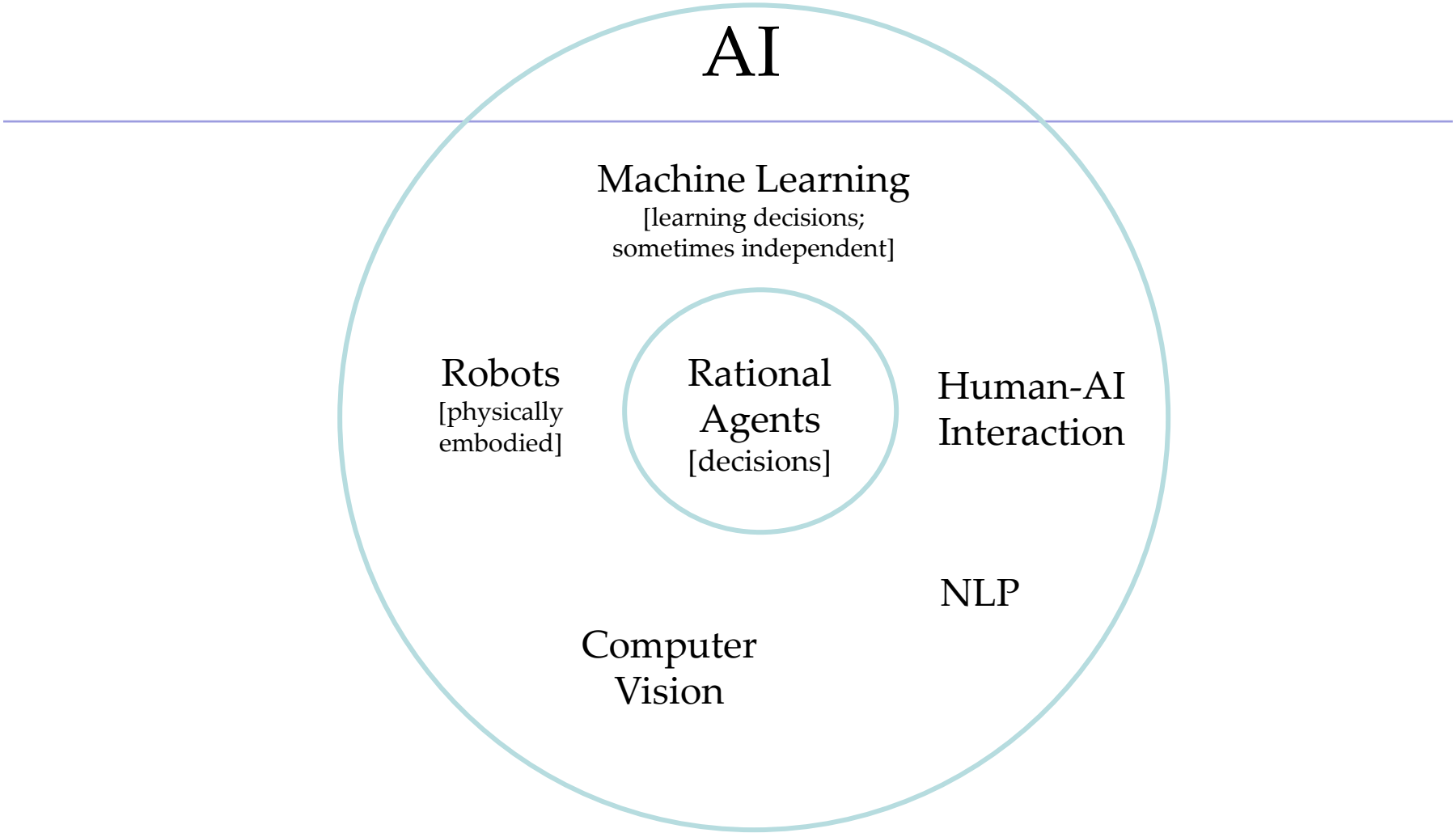
Pac-Man is a registered trademark of Namco-Bandai Games, used here for educational purposes



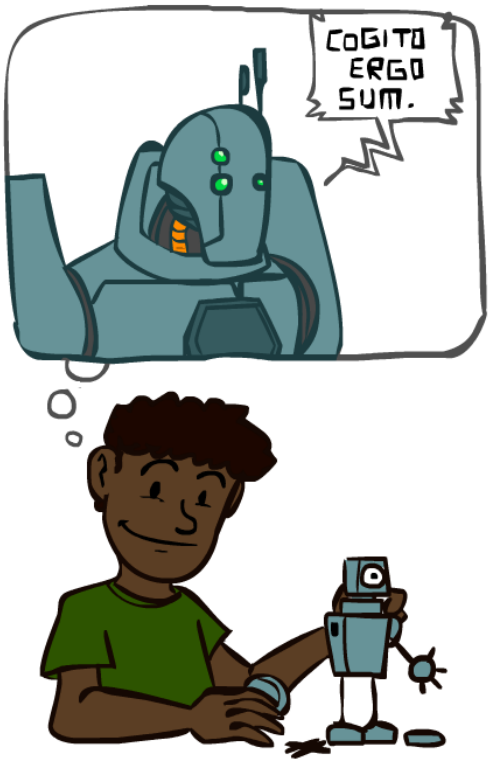
Generalne teme ovog kursa

- Donošenje odluka
  - Pretrage / planiranje
  - Planiranje u okruženju koje sadrži protivnike
  - Planiranje u okruženju koje je stohastičko
  - Planiranje u okruženju o kome agent ne zna ništa (*reinforcement learning*)
- Mašinsko učenje
  - Linearni i nelinearni klasfikatori
  - Klasterovanje
  - Probabilistički modeli
  - Neuronske mreže
  - *Deep Learning...*



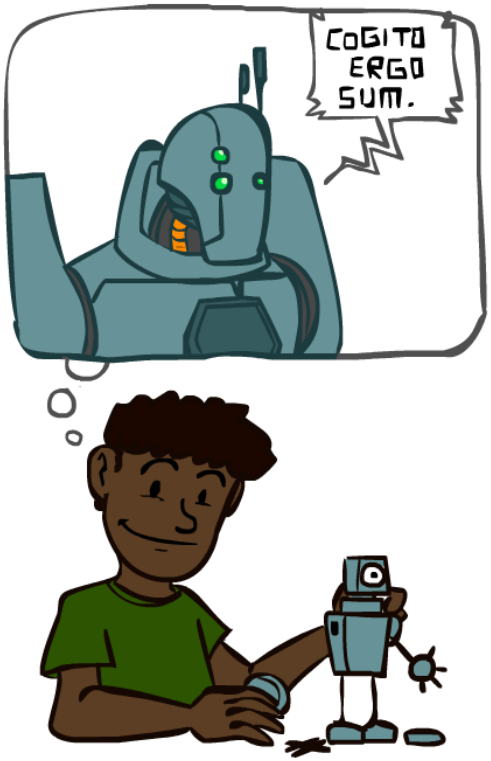

AI
Machine Learning
[learning decisions;
sometimes independent]
Robots
[physically
embodied]
Rational
Agents
[decisions]
Human-AI
Interaction
NLP
Computer
Vision

(Kratka) Istorija AI



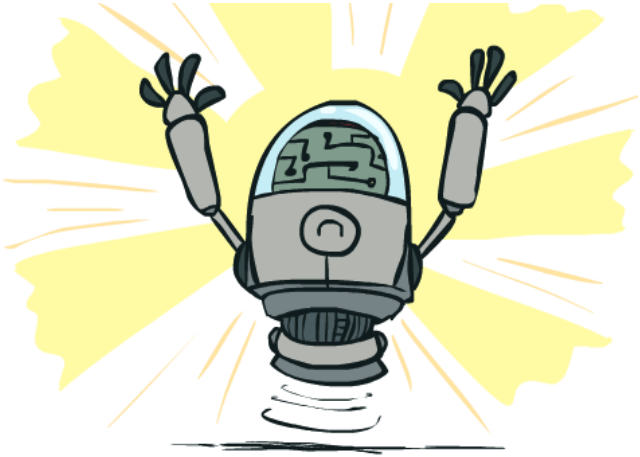
(Kratka) Istorija AI

- 1940—1950: Početak
  - 1943: McCulloch & Pitts: Boolean električno kolo kao model mozga
  - 1950: Turingov rad "Computing Machinery and Intelligence"
- 1950—70: Entuzijazam (oduševljenje):
  - 1950s: Prvi AI programi, AI igra mice (Samuel), AI se bavi matematikom (Newell & Simon's Logic Theorist, Gelernter's Geometry Engine)
  - 1956: Usvojen termin „Veštačka Inteligencija" - ("Artificial Intelligence, AI")
  - 1965: Algoritam za logičko rezonovanje (Robinson)
- 1970—90: Pristupi zasnovani na zanju (*knowledge-based approaches*)
  - 1969—79: Prvi ekspertski sistemi
  - 1980—88: Ekspanzija ekspertskih sistema u industriji
  - 1988—93: Propast ekspertskih sistema u industriji, „Zima za AI" - "AI Winter"
- 1990—2012: Statistički pristupi
  - Fokus na modelovanju nesigurnosti pomoću verovatnoće
  - Generalno povećanje tehničke kompleksnosti AI tehnika
  - Agenti i sistemi koji uče… "AI Proleće"?
- 2012— : Entuzijazam ponovo
  - Puno dostupnih podataka, mnogo bolji hardver, pojava dubokih neuronskih mreža (*deep learning*)
  - AI ima dosta realnih upotreba u industriji



Šta od navednog AI može uspešno da uradi?

Šta AI trenutno može da uradi?



- ✓ Uspešno se takmiči u kvizovima (npr. *Jeopardy*)?
- ✓ Pobedi čoveka u šahu?
- ✓ Pobedi najbolje igrače u igri Go?
- ✓ Igra stoni tenis relativno dobro?
- ✓ Uzme određeni predmet i postavi ga na policu?
- ✗ Izvadi sudove iz bilo koje sudo-mašine u bilo kojoj kući?
- ? Vozi bezbedno auto-putem?
- ✗ Vozi bezbedno kroz Beograd?
- ✓ Na internetu kupi namirnice potrebne za jednu nedelju?
- ✗ Ode u prodavnicu i kupi namirnice potrebne za jednu nedelju?
- ? Otkrije i dokaže novu matematičku teoremu?
- ? Izvrši (uspešno) hiruršku operaciju?
- ✓ Prevodi kineski govor na engleski u realnom vremenu?
- ✗ Napiše priču sa namerom da bude smešna?

AI može da napiše scenario za film koji je nenamreno smešan



Movie written by algorithm turns out to be hilarious and intense

For *Sunspring*'s exclusive debut on Ars, we talked to the filmmakers about collaborating with an AI.

ANNALEE NEWITZ - 6/9/2016, 12:30 PM

Sunspring, a short science fiction movie written entirely by AI, debuts exclusively on Ars today.

Ars is excited to be hosting this online debut of *Sunspring*, a short science fiction film that's not entirely what it seems. It's about three people living in a weird future, possibly on a space station,

https://arstechnica.com/gaming/2016/06/an-ai-wrote-this-movie-and-its-strangely-moving/

AI može da napiše scenario za film koji je nenamreno smešan

- „He is standing in the stars and sitting on the floor. He takes a seat on the counter and pulls the camera over to his back. He stares at it. He is on the phone. He cuts the shotgun from the edge of the room and puts it in his mouth. He sees a black hole in the floor leading to the man on the roof.“

He picks up a light screen and fights the security force of the particles of a transmission on his face.
H
(continuing)
What do you mean?
C
(smiles)
I don't know anything about any of this.
H
(to Hauk, taking his eyes from his mouth)
Then what?

H2
There's no answer.
C
(frowning)
We're going to see the money.
H
(reading)
"All right, you can't tell me that."

ChatGPT: par pokušaja namerni smešnih priča

KO

Write an intentionally funny short story about computer science students.

Once upon a time, in a computer science department of a prestigious university, there were three students named Jack, Jill, and Jerry. Jack and Jill were top students, always getting perfect grades and impressing their professors. But Jerry, on the other hand, was a bit of a slacker. He often slept through lectures and never did his homework.

One day, the professor assigned a group project that would determine half of the grade for the semester. Jack and Jill were paired up, and to their dismay, Jerry was added to their group. They were certain that they would fail the project with Jerry on their team.

But to their surprise, Jerry had a hidden talent, he was a master at making computer programs talk to each other. Jack and Jill had never seen anything like it. With Jerry's help, the group was able to finish the project in record time and it even earned them extra credit from the professor.

Feeling grateful, Jack and Jill asked Jerry how he was able to do it. Jerry simply grinned and said, "I may have slept through lectures, but I spent all that time playing video games, and it taught me how to make the characters talk to each other."

From that day on, Jack, Jill, and Jerry became the most unlikely of friends, working together and acing every computer science assignment they were given. The end.

ChatGPT: par pokušaja namerni smešnih priča

KO

Add a ironic twist to the previous story.

Once upon a time, in a computer science department of a prestigious university, there were three students named Jack, Jill, and Jerry. Jack and Jill were top students, always getting perfect grades and impressing their professors. But Jerry, on the other hand, was a bit of a slacker. He often slept through lectures, never did his homework and was known for his love of memes and cat videos.

One day, the professor assigned a group project that would determine half of the grade for the semester. Jack and Jill were paired up, and to their dismay, Jerry was added to their group. They were certain that they would fail the project with Jerry on their team.

But to their surprise, Jerry had a hidden talent, he was a master at making computer programs talk to each other using memes and cat videos. Jack and Jill had never seen anything like it. With Jerry's help, the group was able to finish the project in record time and it even earned them extra credit from the professor. The professor was so impressed that he even asked Jerry to teach him how to use memes in his lectures.

Feeling grateful, Jack and Jill asked Jerry how he was able to do it. Jerry simply grinned and said, "I may have slept through lectures, but I spent all that time playing video games and browsing memes, and it taught me how to make the programs communicate in a way that even cats can understand."

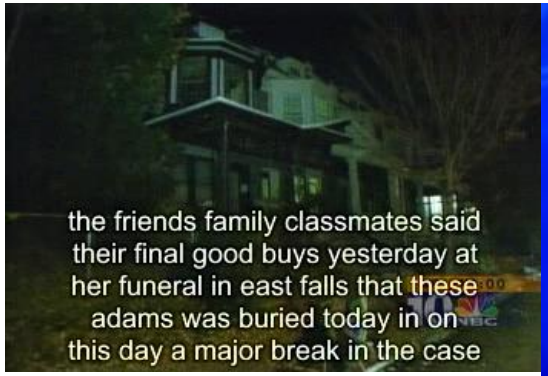
From that day on, Jack, Jill, and Jerry became the most unlikely of friends, working together and acing every computer science assignment they were given. They were even invited to give a talk at a conference about the benefits of using memes in computer science education.

But, as it turned out, the conference was a trap set by their professor, who was actually a robot from the future sent to prevent the use of memes in computer science. The three students were arrested and charged with "meme-crime" and their future in the computer science field was forever tarnished.

The end.

AI i Prirodni Jezik (*Natural Language*)

- Obrada govora (npr. Siri)
  - Prepoznavanje govora (ASR)
  - Sinteza govora od teksta (TTS)
  - Sistemi za dijalog





- Obrada prirodnog jezika (NLP)
  - Odgovaranje na pitanja
  - Mašinski prevod


"Il est impossible aux journalistes de rentrer dans les régions tibétaines"

Bruno Philip, correspondant du "Monde" en Chine, estime que les journalistes de l'AFP qui ont été expulsés de la province tibétaine du Qinghai "n'étaient pas dans l'illégalité".

Les faits Le dalaï-lama dénonce l'"enfer" imposé au Tibet depuis sa fuite, en 1959
Vidéo Anniversaire de la rébellion


"It is impossible for journalists to enter Tibetan areas"

Philip Bruno, correspondent for "World" in China, said that journalists of the AFP who have been deported from the Tibetan province of Qinghai "were not illegal."

Facts The Dalai Lama denounces the "hell" imposed since he fled Tibet in 1959
Video Anniversary of the Tibetan rebellion: China on guard

  - Pretraživanje Weba
  - Klasifikacija teksta, filtriranje spama, etc…

Obrada slike i videa (*Computer Vision*)



"man in black shirt is playing guitar."



"construction worker in orange safety vest is working on road."



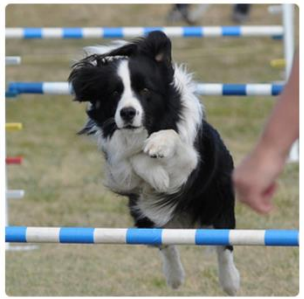
"two young girls are playing with lego toy."



"boy is doing backflip on wakeboard."



"girl in pink dress is jumping in air."



"black and white dog jumps over bar."



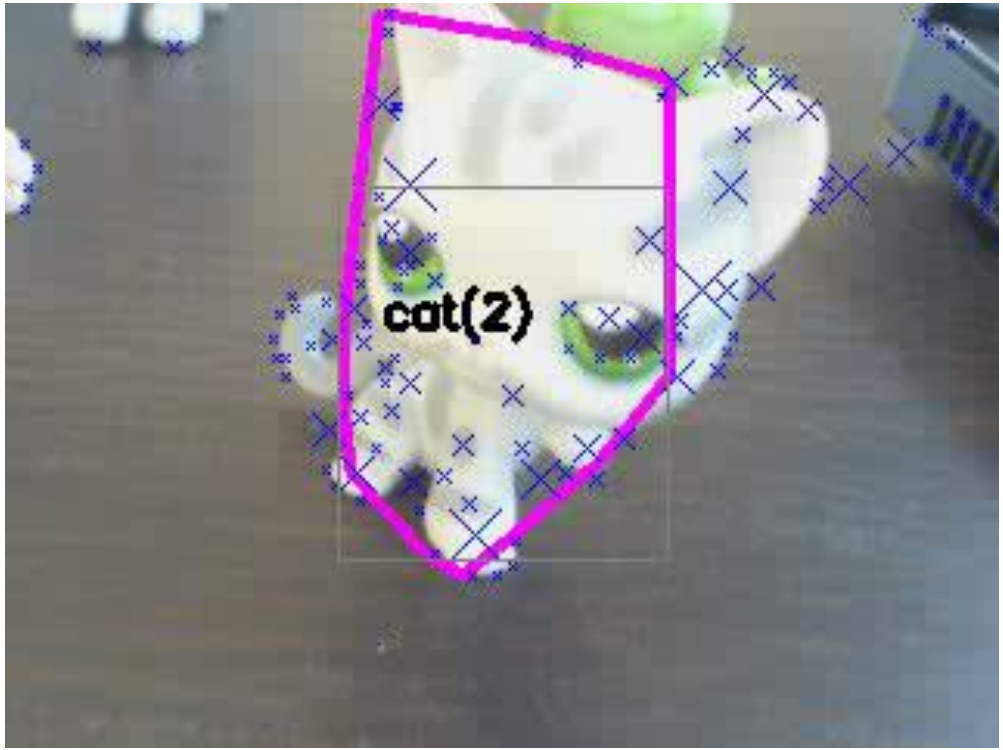
"young girl in pink shirt is swinging on swing."



"man in blue wetsuit is surfing on wave."

Karpathy & Fei-Fei, 2015; Donahue et al., 2015; Xu et al, 2015; many more

Obrada slike i videa (*Computer Vision*)




. e4 c6 2. d4 d5 3. Nc3 dxe4

AI igra igre

- Značajan momenat: Maj, '97: Deep Blue vs. Kasparov
  - AI uz dobar hardver pobeđuje svetskog šampiona u šahu
  - AI igra inteligentno i kreativno.
  - Deep Blue obrađuje 200 miliona pozcija u sekundi.
  - Ljudi su mogli da razumeju razloge za 99.9% poteza koje je pravio Deep Blue.
  - Slične performanse se sada mogu dobiti sa klasterom PC računara.



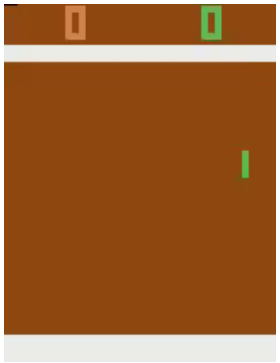
- 1996: Kasparov pobeđuje Deep Blue
  "I could feel --- I could smell --- a new kind of intelligence across the table."
- 1997: Deep Blue pobeđuje Kasparova
  "Deep Blue hasn't proven anything."



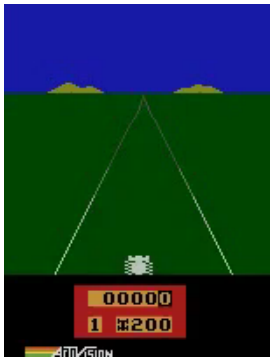
Text from Bart Selman, image from IBM's Deep Blue pages

AI igra igre

- Reinforcement learning (tačnije Deep RL)



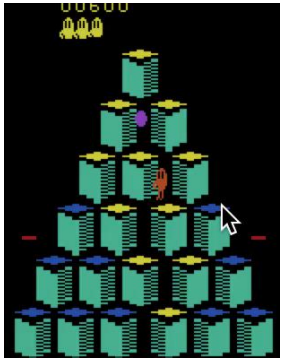
Pong



Enduro



Beamrider



Q*bert



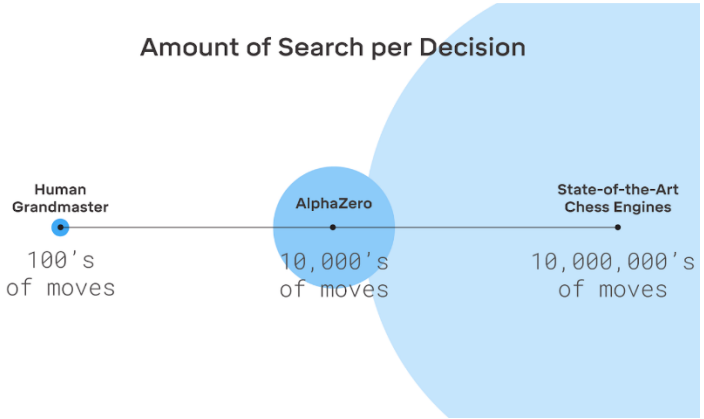
DeepMind: AlphaZero

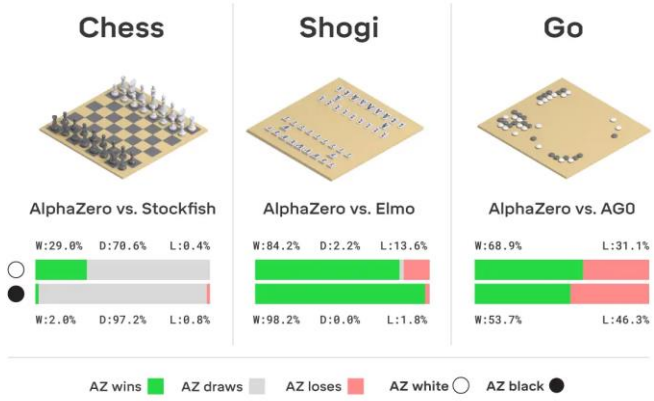
BLOG POST
RESEARCH

06 DEC 2018

AlphaZero: Shedding new light on chess, shogi, and Go







https://deepmind.com/blog/article/alphazero-shedding-new-light-grand-games-chess-shogi-and-go

DeepMind: AlphaZero

> Some of its moves, such as moving the King to the centre of the board, go against shogi theory and - from a human perspective - seem to put AlphaZero in a perilous position. But incredibly it remains in control of the board. Its unique playing style shows us that there are new possibilities for the game."

YOSHIHARU HABU
9-DAN PROFESSIONAL, ONLY PLAYER IN HISTORY TO HOLD ALL SEVEN MAJOR SHOGI TITLES

> The implications go far beyond my beloved chessboard... Not only do these self-taught expert machines perform incredibly well, but we can actually learn from the new knowledge they produce."

GARRY KASPAROV
FORMER WORLD CHESS CHAMPION

https://deepmind.com/blog/article/alphazero-shedding-new-light-grand-games-chess-shogi-and-go

DeepMind: AlphaStar

GAMING | TECH | ARTIFICIAL INTELLIGENCE

DeepMind's StarCraft 2 AI is now better than 99.8 percent of all human players

AlphaStar is now grandmaster level in the real-time strategy game

By Nick Statt | @nickstatt | Oct 30, 2019, 2:00pm EDT

SHARE



Image: DeepMind

https://www.theverge.com/2019/10/30/20939147/deepmind-google-alphastar-starcraft-2-research-grandmaster-level

OpenAI Five

OpenAI's Dota 2 AI steamrolls world champion e-sports team with back-to-back victories

The International 2018 champion OG loses to OpenAI's bots in a stunning defeat

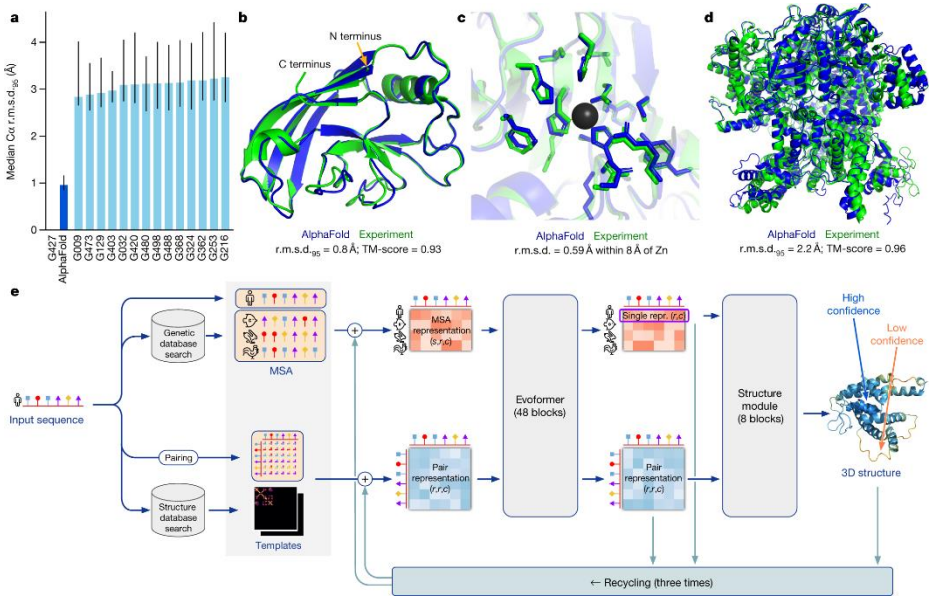
By Nick Statt | @nickstatt | Apr 13, 2019, 5:05pm EDT

SHARE



https://www.theverge.com/2019/4/13/18309459/openai-five-dota-2-finals-ai-bot-competition-og-e-sports-the-international-champion

DeepMind: AlphaFold



https://media.springernature.com/full/springer-static/image/art%3A10.1038%2Fs41586-021-03819-2/MediaObjects/41586_2021_3819_Fig1_HTML.png?as=webp

AI Reduces Timescale of Early Drug Development to Weeks

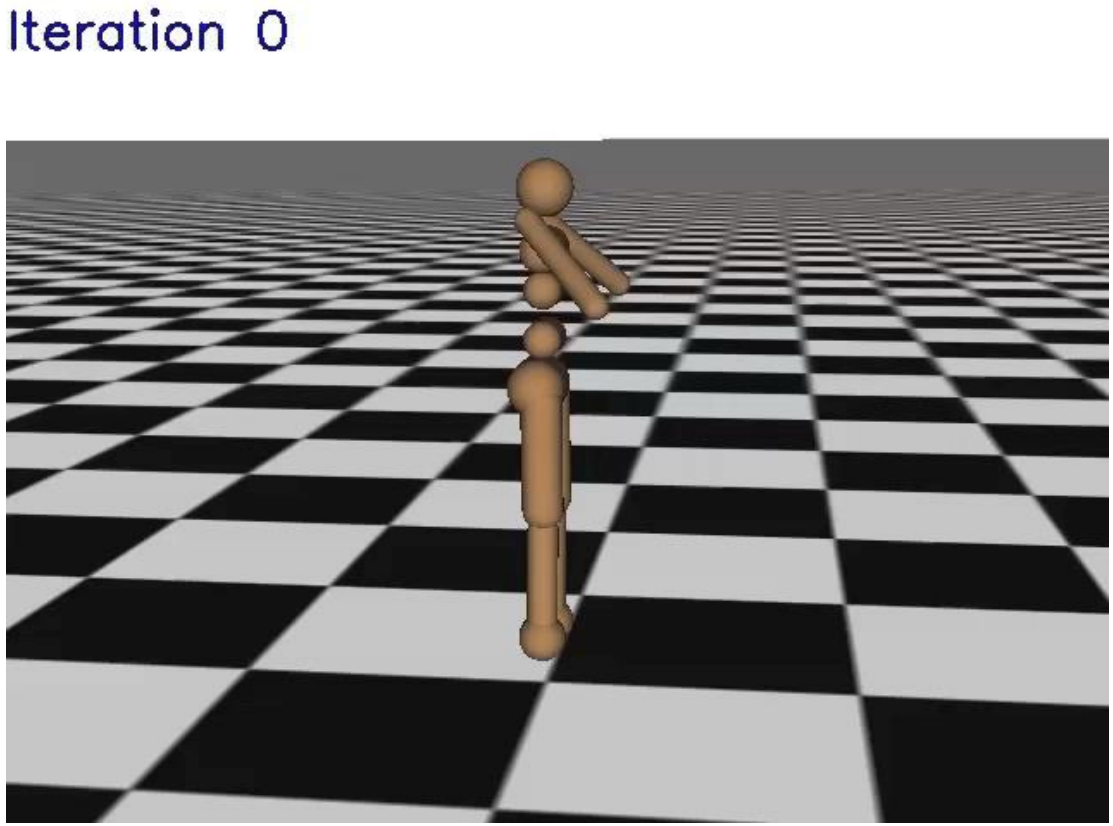
The process of finding potential drugs for human diseases is being revolutionized by AI discovery engines.

The Physics arXiv Blog | By The Physics arXiv Blog | Jan 28, 2022 5:41 PM



https://www.discovermagazine.com/the-sciences/ai-reduces-timescale-of-early-drug-development-to-weeks

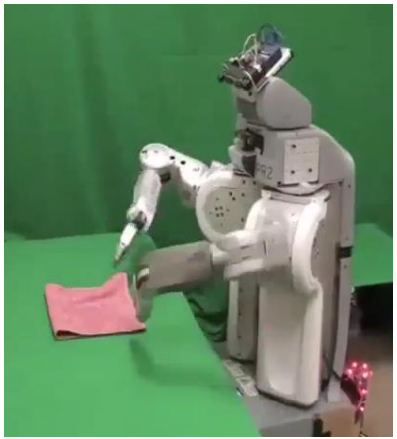
AI uči da hoda u simulaciji



[Schulman, Moritz, Levine, Jordan, Abbeel, ICLR 2016]

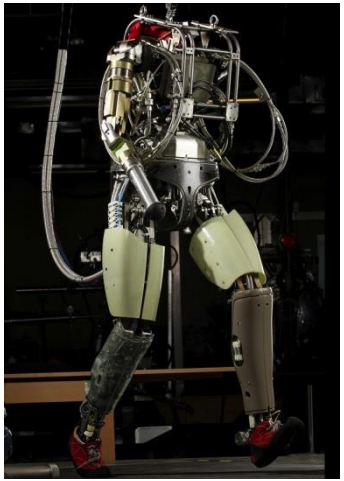
Robotika

- Robotika
  - Delom mehanika,
  - Delom AI
  - Rad u realnosti je mnogo teži od simulacija!
- Tehnologije
  - Vozila
  - Spasavanje (npr. robot demontira bombe)
  - Pomoć u kući
  - …
- Na ovom kursu:
  - Ignorišemo mehaničke aspekte
  - Bavimo se metodama za planiranje i kontrolu.



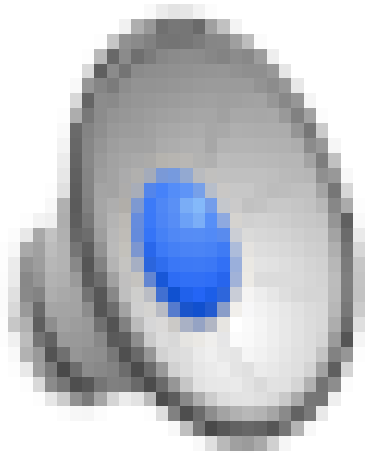






Images from UC Berkeley, Boston Dynamics, RoboCup, Google

Robots



[Levine*, Finn*, Darrell, Abbeel, JMLR 2016]

Robotika



Robotika



Vrhunac robotike u 2023 ☺

Marines fooled a DARPA robot by hiding in a cardboard box while giggling and pretending to be trees

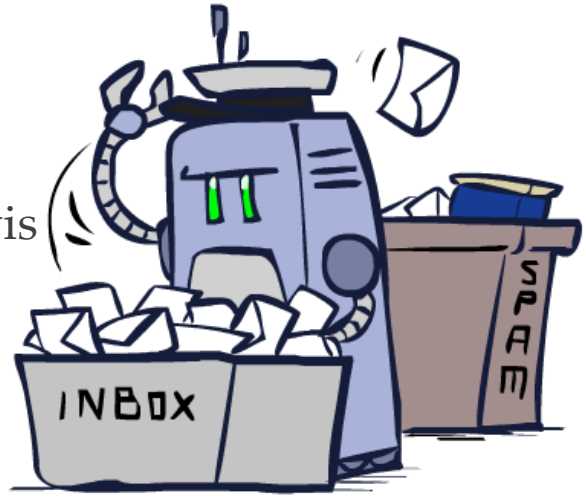
Katherine Tangalakis-Lippert Jan 21, 2023, 8:11 AM



Donošenje odluka



- Primenjena veštačka inteligencija automatizuje mnoge stvari
  - Raspoređivanje i rutiranje (letovi, vojska...)
  - Određivanje optimalnih putanja npr. Google maps
  - Medicinska dijagnostika
  - Pretraživanje interneta
  - Filtriranje spama
  - Automatomatizovani korisnički servis
  - Detekcija prevara
  - Preporuka proizvoda
  - …



O predmetu

- Predavanja
  - objašnjenja algoritama uz primere
- Vežbe
  - ideja je da sami implemtirate i primenite metode koje učite na predavanjima

Pravila polaganja

- Obaveze na predmetu sastoje se od dva dela:
  - Predispitne obaveze
  - Teorijski ispit

Pravila polaganja - Predispitne obaveze

- Sastoje se od dva kolokvijuma i projekta
  - kolokvijumi će biti održani samo jednom i to u toku semestra

Pravila polaganja - Predispitne obaveze

- Termini za računarske kolokvijume:
  - I kolokvijum,
    - u nedelji od 27.03.2023. do 01.04.2023.
      tačni termini biće vam sapošteni na vreme
  - II kolokvijum,
    - u nedelji od 29.05.2023. do 03.06.2023.
      tačni termini biće vam sapošteni na vreme

Pravila polaganja - Projekat

- Umesto polaganja drugog računarskog kolokvijuma postoji mogućnost izrade projekta
- Izrada projekta ima nekoliko faza
  - Specifikacija i predaja predloga projekata
  - Izrada projekta
  - Odbrana projekta

Pravila polaganja - Projekat

- Specifikacija i predaja predloga projekata
  - U periodu od 03.04.2023. do 11.05.2023. studenti koji žele da rade projekta pišu predlog projekta
  - Studenti treba sami da osmisle temu projekta
  - Okvirne informacije o temama dobićete kroz primere na predavanjima i vežbama i kroz materijale sa sajta predmeta
  - Informacije o tome šta sve treba da sadrži specifikacija dobićete od asistenata
  - U periodu od 11.05. - 25.05. asistenti razmatraju podnete predloge i donose odluke o tome koji projekti mogu da pređu u sledeću fazu
  - Studenti čiji predlozi ne prođu u sledeću fazu moraju da rade drugi računarski kolokvijum

Pravila polaganja - Projekat

- Izrada projekta
  - Od momenta kada asistent odobri projekat studenti počinju sa izradom projekta
  - Detalje oko načina na koji će asistenti pratiti projekte će vam oni saopštiti kad za to dođe vreme

- Odbrana projekta
  - Postoji samo jedan termin za odbranu projekata
  - Tačan termin će objaviti asistenti i znaće se na vreme
  - Okvirno, odbrana će biti sredinom jula
  - Način odbrane će takođe objaviti asistenti

Pravila polaganja - Projekat

- Važna napomena:
  - Kada student pristupi izradi projekta tj. kada je asistent odobrio projekat:
    - Student nema pravo polaganja drugog računarskog kolokvijuma
    - Dakle imate izbor, ali morate sami da donesete odluku šta želite da uradite

Pravila polaganja - Teorijski ispit

- Teorijski ispit
  - polaže se pismeno u terminima rasporeda ispita koje određuje fakultet
  - sastoji se od 2 dela koji se mogu polagati odvojeno ili zajedno

Bodovanje – Teorijski deo

Bodovanje teorijskog dela ispita:

	min	max
Deo 1	50	100
Deo 2	50	100
Ukupno	100	200
Ukupno (preračunato)*	26	45

$$* \textit{Ukupno teorijski ispit} = \frac{(Deo1 + Deo2)}{200} * 38 + 7$$

Bodovanje – Predispitne obaveze

Bodovanje predispitnih obaveza:

	min	max	max sa projektom
Kolokvijum 1	13	25	25
Kolokvijum 2	12	25	0
Projekat *	0	0	35
Ukupno	25	50	60

*Projekat može birati svaki student i radi se umesto drugog kolokvijuma.

Važna napomena: Morate osvojiti bar 13 bodova na prvom i 12 na drugom kolokvijumu da bi položili predispitne obaveze.

Ukupno:

	min	max	max sa projektom
Ukupno teorija	26	45	45
Ukupno praktični	25	50	60
Ukupno	51	95	105

Zaključna ocena bez teorijskog dela ispita

- Ako ste položili predispitne obaveze stičete pravo na zaključnu ocenu bez polaganja teorijskog dela.

- Bodovi sa predispitnih obaveza se na zaključnu ocenu preslikavaju na sledeći način:

[50, 45) ocena 10
[45, 40) ocena 9
[40, 35) ocena 8
[35, 30) ocena 7
[30, 25] ocena 6

Zaključna ocena bez teorijskog dela ispita

- Ako niste zadovoljni ocenom bez teorijskog dela:

1. Onda polažete teorijski deo ispita u zvaničnim terminma koje određuje FTN.

2. Morate da doložite oba dela teorije
   - Da osvojite minimalno 50 bodova iz svakog dela.

3. Zaključna ocena računa se po pravilima sa ovih slajdova, odnosno gubite pravo na zaključnu ocenu bez teorijskog dela.

Zaključna ocena bez teorijskog dela ispita

- Odluku o tome da li ćete prihavtiti ocenu bez teorijskog dela možete doneti do početka sledeće školske godine.
- Prvim izlaskom na teorijski deo ispita gubite pravo na ocenu bez teorijskog dela.
- Sve navedeno važi i za studente koji će predispitne obaveze položiti nakon popravnog računarskog kolokvijuma ili nakom odbrane projekta.

Zaključna ocena bez teorijskog dela ispita
Važna napomena

- Da biste imali pravo na zaključnu ocenu bez teorijskog dela, obavezno je da:

1. **Položite prvi računarski kolokvijum**
2. **Položite drugi računarski kolokvijum ili odbranite projekat**

Polaganje ispita sa teorijskim delom
Važna napomena

- Da biste imali pravo na zaključnu ocenu sa teorijskim delom ispita, obavezno je da:

1. **Položite oba dela teorije**
   - Da osvojite minimalno 50 bodova iz svakog dela
2. **Položite prvi računarski kolokvijum**
3. **Položite drugi računarski kolokvijum ili odbranite projekat**

Literatura

- Teorijski deo
  - Slajdovi sa predavanja
  - Prateći materijali koji će biti postavljeni na sajt predmeta
- Predispitne obaveze
  - Materijali sa vežbi

Literatura

- Ova knjiga nije nepohodna za polaganje predmeta, ali je preporuka za one od vas kojima su AI teme zanimljive i koji žele da nauče više.
- Napomena: Slajdovi ne prate ovu knjigu, ali knjiga sadrži materijale vezane za dobar deo obasti koje radimo.
  - Russell & Norvig, AI: A Modern Approach, 3rd Ed.

